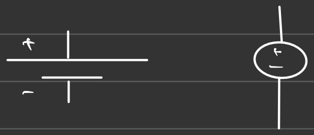
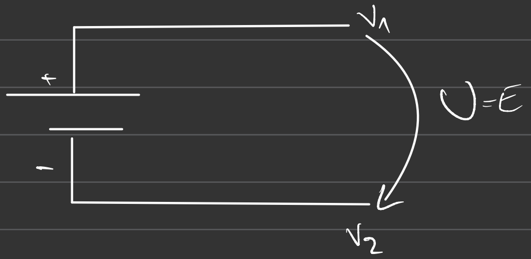
Curs 1



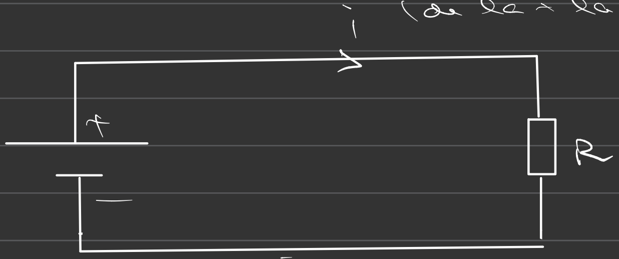
sursă de curent

E=5V



Diferenta de potential =
Tensiune =
Cădere de tensiune

$U = V_1 - V_2$

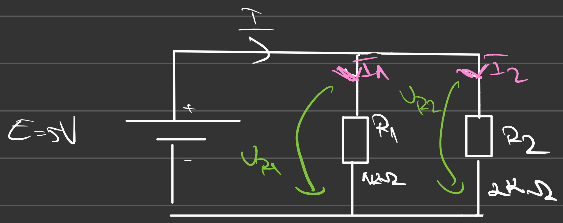


i (de la + la - : intensitatea)

$\bar{e} \rightarrow$ (de la - la +)

Legea lui Ohm: $\boxed{U = R \cdot I}$

$I = \frac{U}{R} = \frac{E}{R}$

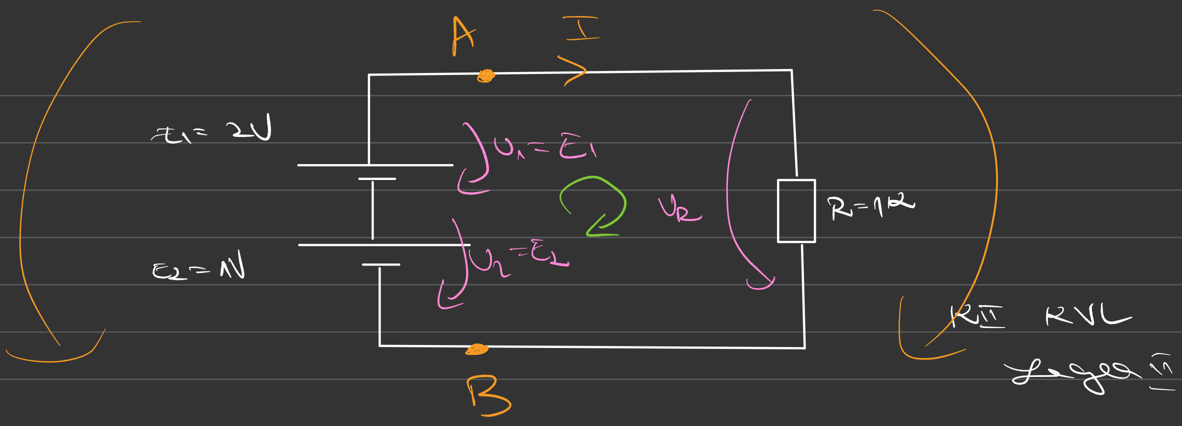


$U_{R1} = E = U_{R2}$

$I = I_1 + I_2$ KI KCL (Legea lui Kirchhoff)

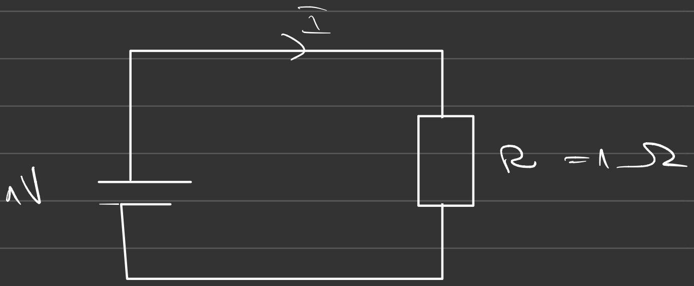
$I - I_1 - I_2 = 0$

$$\left.\begin{array}{l} I_1 = \frac{U}{R_1} = 5mA \\ I_2 = \frac{U}{R_2} = 2,5mA \end{array}\right\} \Rightarrow I = 7,5mA$$



$-U_2 - U_1 + U_R = 0$

Dacă trec prin cele 2 puncte pe st ↓ sau pe dr ↓ trebuie să obținem același rezultat

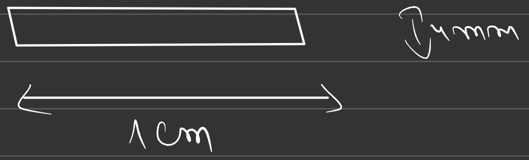


$I = 1A = 1\frac{C}{s}$ (Coulomb/secundă)

$e_0 = 1.6 \cdot 10^{-19} C \simeq 10^{19}\ \bar{e}/s$

Pentru $I = 1A$ trec $\simeq 10^{19}\ \bar{e}$ pe secundă

Cu $\quad d = 4\,\mathrm{mm}$



$$V = (2\,\mathrm{mm})^2 \cdot \pi \cdot 1\,\mathrm{cm} = 10\,\mathrm{mm}^2 \cdot 1\,\mathrm{cm} =$$
$$= 0{,}1\,\mathrm{cm}^3$$

Numărul lui Avogadro (NA)

1 mol Cu $\qquad$ Cu = 60g $= 6 \cdot 10^{23}$ atomi

$$\mathrm{Cu} = \frac{6 \cdot 10^{23}}{60} = 10^{22}\ \text{atomi/g}$$

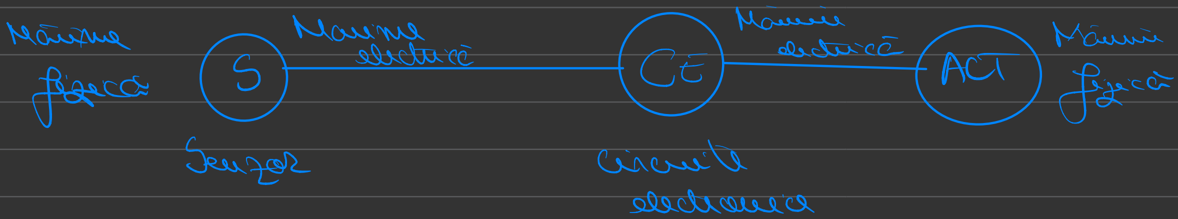
1cm cablu $= 0{,}1\,\mathrm{cm}^3 \cdot 6\,8/\mathrm{cm}^3 \simeq 1\,\mathrm{g}$

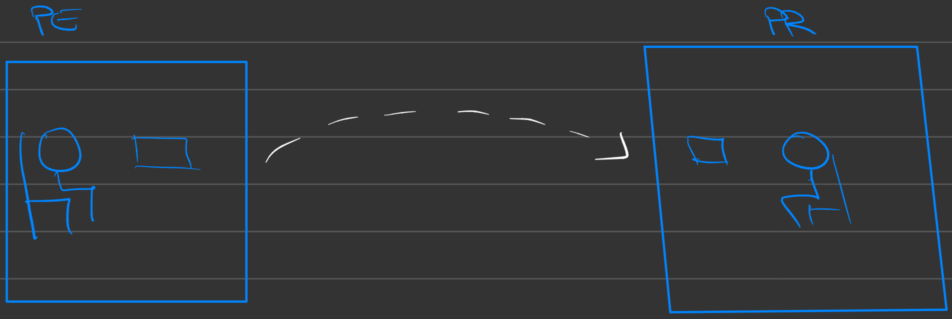
$\Leftrightarrow 10^{22}$ atomi/cm de cablu

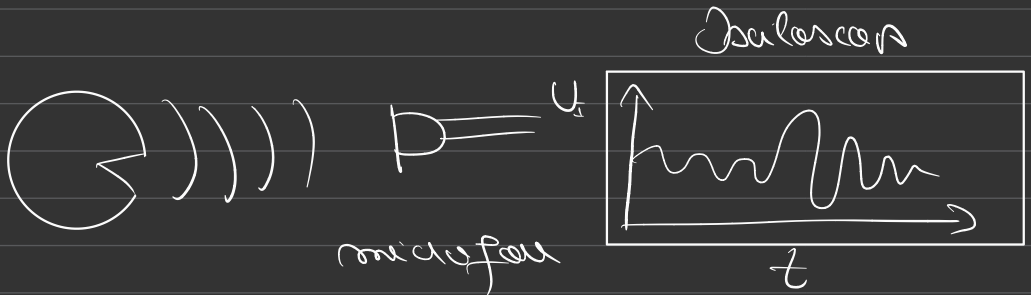
$\Leftrightarrow 2 \cdot 10^{22}$ $\bar{e}$/cm de cablu

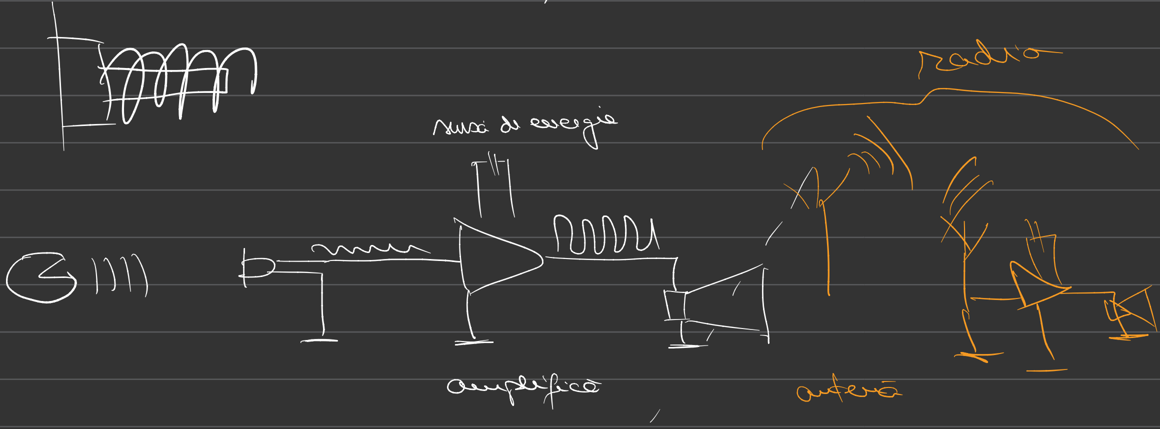
$$V = \frac{10^{19}\,\frac{e}{s}}{2 \cdot 10^{22}\,\frac{e}{\mathrm{cm}}} = \frac{1}{2000} = 0{,}0005\ \mathrm{cm/s}$$
$$\simeq 2\ \mathrm{cm/h}$$

(Viteza cu
care se deplasea-
ză un $\bar{e}$
în cablu)


Mărime fizică
S
Senzor
Mărime electrică
CE
Circuite electronice
Mărime electrică
ACT
Mărime fizică


PE
PR





Electronică analogică

MF ≃ ME (sunt proporționale)